CRAIG TANAKA CHIPENDO

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PROFESSIONAL SUMMARY

Data Scientist with 2+ years shipping models and AI systems into production. Tech lead on a 5-person team that built a multi-agent RAG system on Vertex AI: owned the architecture, built and integrated every agent, shipped the first version of the RAG, and built the evaluation framework that scored output across **six metrics** against verified ground truth, lifting **Recall@5 from 85.7% to 92.9%**. Built LSTM forecasting and anomaly detection at **0.89 AUC-ROC** and **91% precision**, used by an engineering department for early warning ahead of outages, and a credit risk model at **0.84 AUC-ROC** running in a production reporting system. Proves what shipped worked, with geo holdout designs and drift monitoring. MS Business Analytics, Emory (2026); BSc Data Science, GPA 4.0, top of class.

EDUCATION

Emory University, Goizueta Business School *May 2026*
Master of Science, Business Analytics *Atlanta, GA*

- Merit Scholarship Recipient and Graduate Research Assistant, Data Science. Coursework: AI and Machine Learning, Deep and Reinforcement Learning, Causal Inference, Optimization (Gurobi, OR-Tools), Managing Big Data.

University of Zimbabwe *Jun 2024*
Bachelor of Science, Data Science and Systems (GPA: 4.0) *Harare, Zimbabwe*

- University Book Prize, Best Student in Class of 2024, ranked first in the graduating class. Data Science Zimbabwe Ambassador, presenting on machine learning deployment at the Big Data Conference, 2022.

TECHNICAL CAPABILITIES

- **Machine Learning and Statistics:** Classification, Regression, Segmentation, Class Imbalance Handling (SMOTE), Hyperparameter Tuning (GridSearchCV), Regularization (Lasso and Ridge), Dimensionality Reduction (PCA), Causal Inference, A/B Testing
- **Generative and Agentic AI:** Large Language Models, Prompt Engineering, Tool Calling, Guardrails, Context Management, Structured Outputs, Embeddings, Chunking Strategies, RAGAS, LangChain
- **Engineering, Platforms and Delivery:** Pandas, NumPy, Scikit-learn, R, FastAPI, SQLAlchemy, PostgreSQL, MySQL, MongoDB, Vector Databases (pgvector, ChromaDB), Amazon S3 and EMR, Apache Hive, Parquet, Docker, Kubernetes, Git and GitHub, CI/CD, Tableau, Seaborn, ggplot2, Methodology Documentation. Certifications: Agentic AI, RAG, Google Cloud Digital Leader

WORK EXPERIENCE

Data Scientist (Practicum) | The Home Depot | Atlanta, GA *Dec 2025 – May 2026*

- **Tech lead on a 5-person team**: designed the architecture and built and integrated every agent on Vertex AI and Google ADK, wiring a delegator, parallel weather and site agents, and a policy agent that validates every recommendation against the RAG layer before release.
- Built the site agent's custom tools over BigQuery order and address-level history with a Google Maps API fallback, and shipped the first version of the RAG, parsing policy document stacks with Gemini Vision into a vector database.
- Benchmarked vector search, BM25, and RRF fusion, then added MMR diversity reranking to lift Recall@5 from 85.7% to 92.9%, and built the evaluation framework scoring **six metrics** with LLM-as-a-judge against ground truth.

Data Analytics and AI Specialist | Liquid Intelligent Technologies | Harare, Zimbabwe *Mar 2025 – Jul 2025*

- Built and deployed LSTM models in TensorFlow for forecasting and anomaly detection at **0.89 AUC-ROC** and **91% precision**, used by the engineering department for early warning ahead of network outages.
- Proved the deployment worked with a geo holdout design, comparing live regions against regions not yet deployed, then engineered drift detection and automated retraining that cut **detection latency by 30%**.
- Built **six executive dashboards** in Power BI across 4 business units, translating model output for non-technical leadership and informing **$2M+** in investment decisions.

Data Analytics Graduate Trainee | Liquid Intelligent Technologies | Harare, Zimbabwe *Sep 2024 – Feb 2025*

- Engineered **7 ETL pipelines** in BigQuery orchestrated with Apache Airflow, running PySpark on Dataproc with validation logic that improved **data quality by 12%** across 4 regulated environments.
- Developed a Flask application with REST APIs integrating **120+ previously disconnected sources** into one interface, cutting average fault identification time by **2 hours**.
- Shipped automated Python monitoring applying statistical pattern analysis to live data, resolving roughly **9 incidents per month 30% faster**.

Data Science Intern | Expert Decision Systems (XDS) | Harare, Zimbabwe *Aug 2022 – Jul 2023*

- Built a credit risk model with weight-of-evidence and interaction feature engineering at **0.84 AUC-ROC** and **11% lift**, integrated into the firm's production reporting system and used by lending teams daily.
- Mined **300M+ records** in SQL and PySpark to surface underwriting and portfolio risk drivers, building the case that led to **3 lending policy revisions**.
- Automated 3 reporting pipelines, cutting turnaround by roughly **1 hour per cycle** and presenting findings to executive stakeholders, driving a **5% uplift** in decision-making efficiency.

PROJECT EXPERIENCE

Fairness-Aware Computer Vision, Honors Thesis | University of Zimbabwe | Harare, Zimbabwe *2024*

- Applied a debiasing variational autoencoder architecture from the published literature in TensorFlow to facial detection, training on CelebA and ImageNet across **400K+ images**.
- Adaptively resampled underrepresented features during training, then evaluated across skin tone and gender subgroups on the Pilot Parliaments Benchmark, cutting the measured bias gap from roughly **28% to 7%** at a debiasing strength of $\alpha = 0.1$.